

section 3.4: Switch

pages 58-59

We won't be concentrating on `switch` statements much (they're a bit of a luxury; there's nothing you can do with a `switch` that you can't do with an `if/else` chain, as in section 3.3 on page 57). But they're quite handy, and good to know about.

The example on page 59 is about as contrived as the example in section 1.6 (page 22) which it replaces, but studying both examples will give you an excellent feel for how a `switch` statement works, what the `if/then` statements are that a `switch` is equivalent to and how to map between the two, and why a switch statement can be convenient.

In the example in the text, note especially the way that ten `case` labels are attached to one set of statements (`ndigit[c-'0']++;`). As the authors point out, this works because of the way switch cases ``fall through," which is a mixed blessing.

The danger of fall-through is illustrated by:

```
switch(food) {
    case APPLE:
        printf("apple\n");

    case ORANGE:
        printf("orange\n");
        break;

    default:
        printf("other\n");
}
```

When `food` is `APPLE`, this code erroneously prints

```
apple
orange
```

because the `break` statement after the `APPLE` case was omitted.

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